PRITHVIRAJ CHAUDHURI

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SUMMARY

Principal Software Applications Engineer with 8 years of experience building enterprise software, cloud-native platforms, and AI-powered applications.

TECHNICAL SKILLS

Programming Languages	- Backend : Python, Java Front End: React, Typescript, JQuery, Javascript.
Frameworks	- LangGraph, Google ADK, Langchain, FastAPI, AEM, Apache Sling, Java Spring Boot, Django
Platforms	- OpenShift, Ansible, Gemini Enterprise, AWS, EKS, OpenEdx, ServiceNow, Google Cloud Platform, Langfuse, MLFlow
Other Skills	- Datastores: MySQL, PostgreSQL, JCR, Redis Gitlab CI/CD pipelines, Github Actions, Helm Charts, Kubernetes, Linux, Bash, APIs, OpenStack

EDUCATION

Master of Computer Science August 2019 – May 2021 GPA – 3.92
North Carolina State University (NCSU), Raleigh NC

Bachelor of Technology, Computer Science and Engineering August 2012 – June 2016 GPA – 8.71/10
RCC Institute of Information Technology (Affiliated to Maulana Abul Kalam Azad University of Technology), Kolkata, India

WORK EXPERIENCE

Principal Software Applications Engineer, Red Hat June 2021 – Present

- Contributed to an enterprise-ready MCP server with DCR authentication and a Redis backend, enabling associates to interact with Google Workspace.
- Built a data pipeline to ingest observability data into Snowflake, improving analysis and availability for enterprise agents.
- Developed features for an AI agent platform, enabling users to build no-code agents with the LangGraph Deep Agent harness.
- Built custom agents for Gemini Enterprise using MCP, A2A, and agent-defined tools.
- Designed Python-based RAG solutions connecting AEM and Google Drive content to Gemini Enterprise through a custom connector.
- Implemented an internal people-search agent using LangGraph, Gemini, and FastAPI.
- Contributed to a platform- and LLM-agnostic RAG framework for internal documentation, using LangChain, Python, and a Quarkus chatbot.
- Customized the AEM sales and marketing platform with React, Java, and JCR to streamline the asset lifecycle.
- Migrated applications and hosts from legacy data centers to OpenStack and OpenShift VMs using Linux images and Ansible playbooks.
- Developed Python and Bash scripts to run asynchronous jobs and support project requirements.
- Diagnosed and resolved issues in multithreaded applications and distributed transactions to improve execution times.
- Designed and developed GitLab CI/CD workflows to deploy application infrastructure and run automated tests on OpenShift.

Graduate Research Assistant, North Carolina State University January 2020 – Present

- Implemented custom components (XBlocks) on OpenEdx for smart course design and migration of courses between instances
- Integrated research projects to enable more intelligent and dynamic course generation in OpenEdx with Java, Python and Django
- Built dashboards with Django and converted apps to container deployments, for consistency of deployment and maintenance

Full Stack Web Developer Intern, Red Hat May 2020 – August 2020

- Used Java Spring Boot and Pug/Jade to implement bulk upload of data to resource management application
- Enhanced search application to enable smart searching of employee records in the organization
- Designed and developed automated deployment driven by Jenkins and Git for faster delivery times and easy maintenance

Systems Engineer, TATA Consultancy Service Ltd, Kolkata India July 2016 – August 2019

- Implemented Service Catalog on ServiceNow for end users to utilize ServiceNow ITSM modules using CMS
- Integrated ServiceNow with external systems to streamline information flow
- Automated tasks by translating manual processes to system and leveraging ServiceNow orchestration platform
- Designed and developed a unified REST API integration application on ServiceNow to enable faster integration
- Migrated Service Catalog hosted in CMS to Service Portal using AngularJS